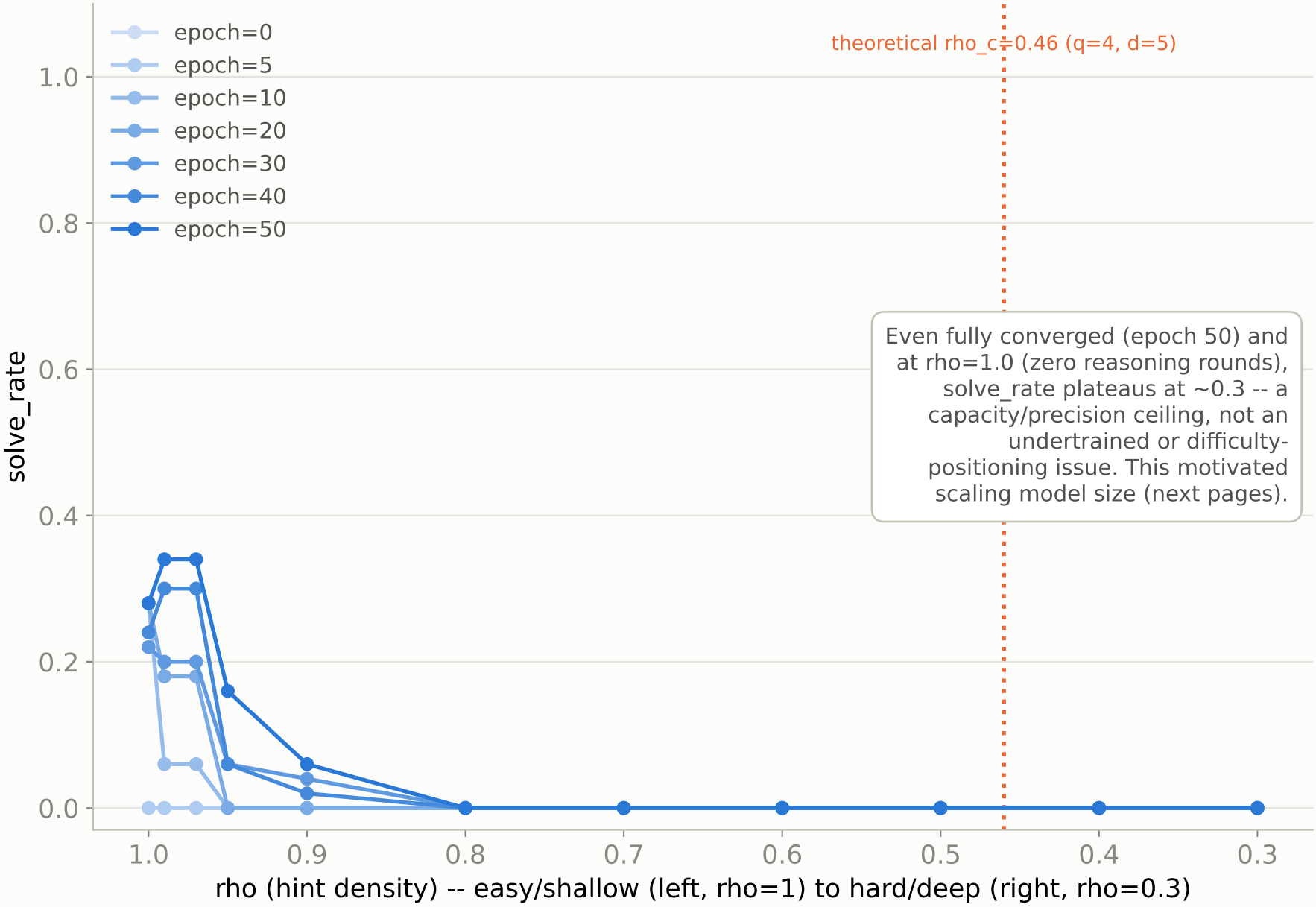
DD-Sudoku transformer: model-size hyperparameter scan

Does scaling model depth/width reach ~100% solve_rate on the near-trivial (O(1) uncolored nodes) end of the task?

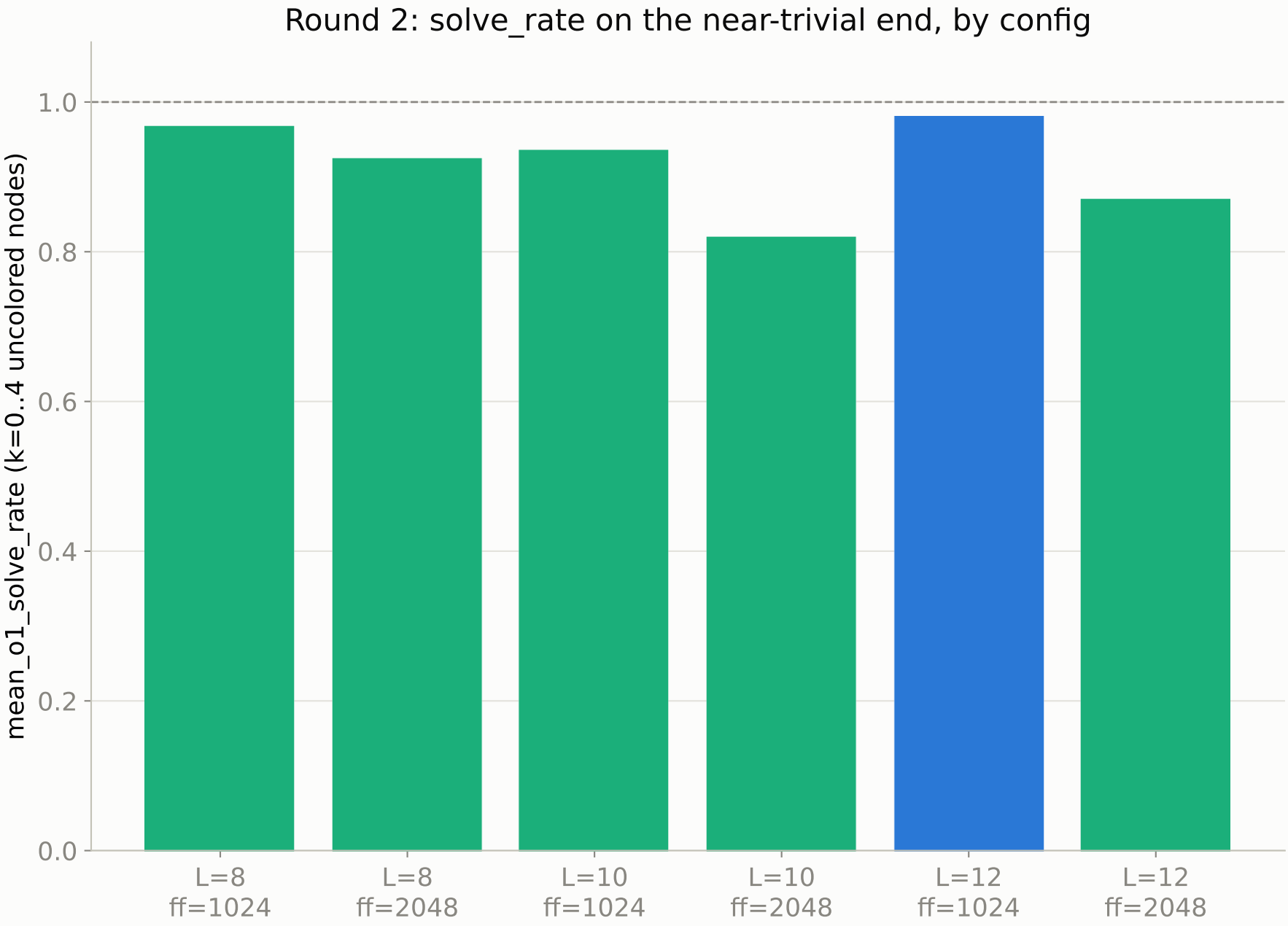
Headline: num_layers=12, dim_feedforward=1024 (d_model=64, nhead=4, 300 epochs) reaches mean_o1_solve_rate = 0.980 across k=0-4 uncolored nodes, up from a 0.480 same-scale baseline and the original pipeline's ~0.01-0.16.

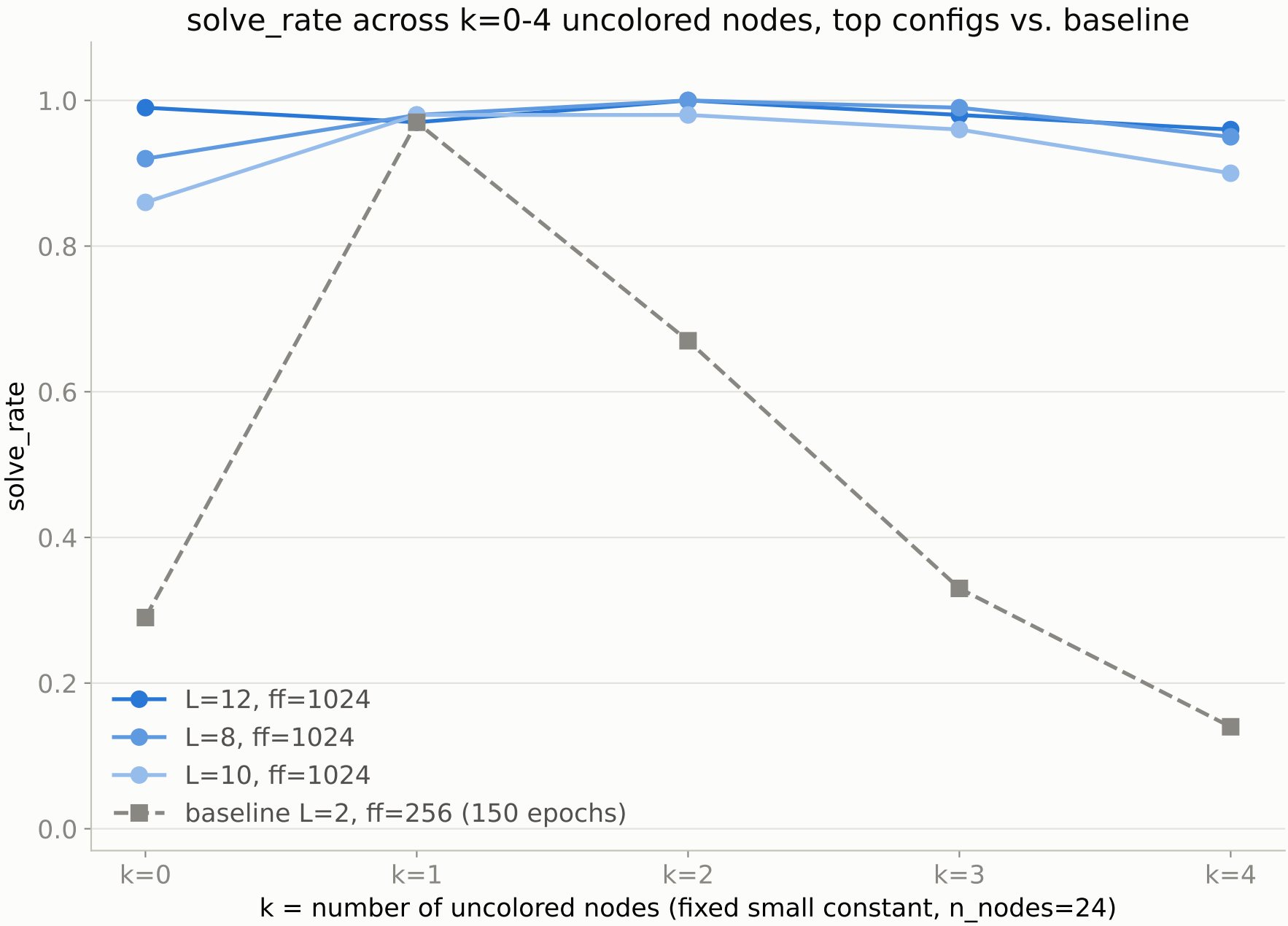
Slurm 48157903 (round 1, 16 configs) + 48162371 (round 2, 6 configs) -- sudoku-dd branch, 2026-10-16

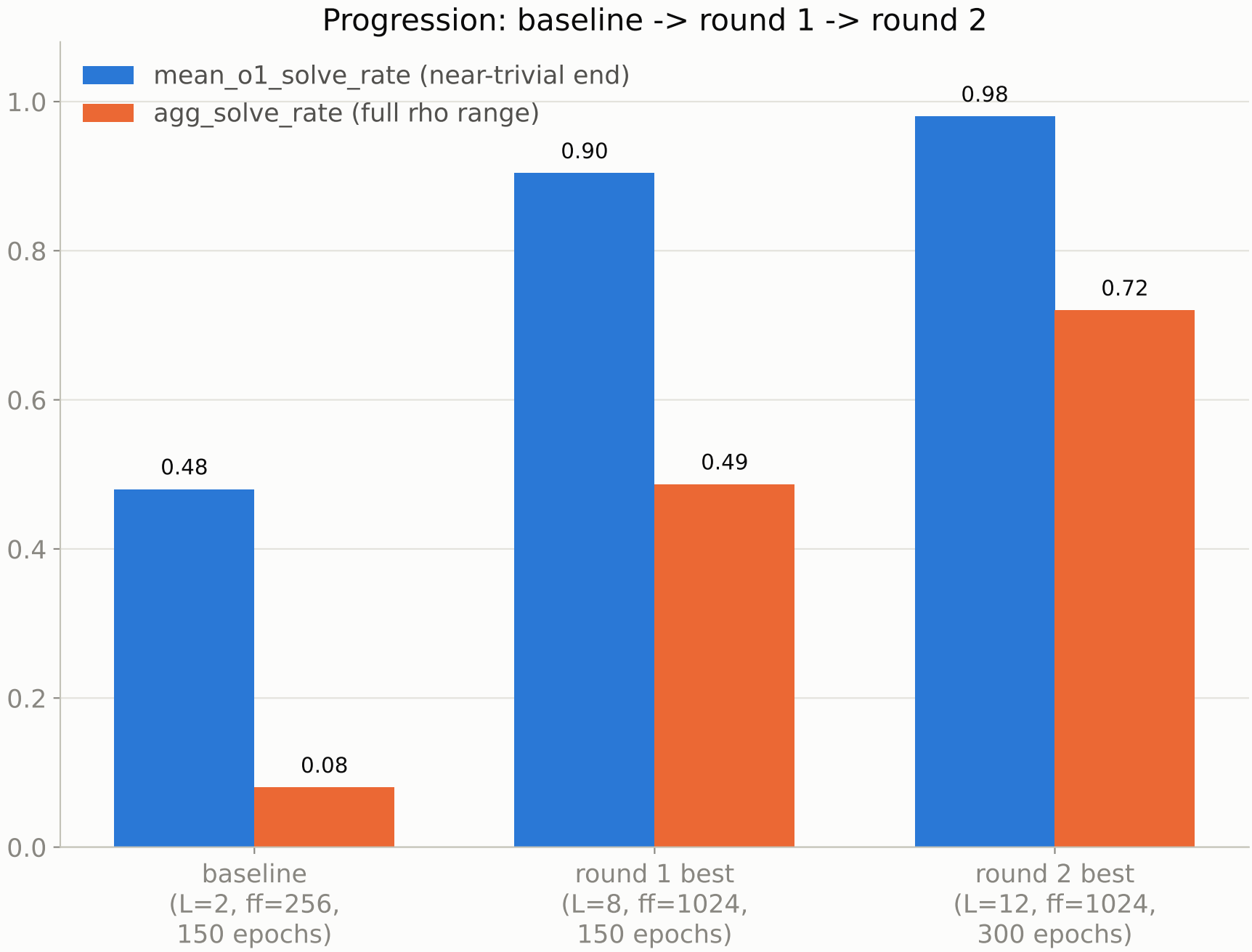

Motivation: baseline (d_model=64, num_layers=2) solve_rate
by checkpoint across the easy-to-hard transition
epoch=0
epoch=5
epoch=10
epoch=20
epoch=30
epoch=40
epoch=50
theoretical rho_c=0.46 (q=4, d=5)
Even fully converged (epoch 50) and at rho=1.0 (zero reasoning rounds), solve_rate plateaus at ~0.3 -- a capacity/precision ceiling, not an undertrained or difficulty-positioning issue. This motivated scaling model size (next pages).
solve_rate
1.0
0.8
0.6
0.4
0.2
0.0
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
rho (hint density) -- easy/shallow (left, rho=1) to hard/deep (right, rho=0.3)

Round 2 results: num_layers x dim_feedforward @ 300 epochs

layers	ffn width	params	mean O(1) solve_rate	agg solve_rate (full range)	val accuracy
12	1024	1,806,626	0.980	0.720	0.575
8	1024	1,210,402	0.968	0.720	0.577
10	1024	1,508,514	0.936	0.623	0.575
8	2048	2,267,170	0.924	0.597	0.566
12	2048	3,391,778	0.870	0.493	0.552
10	2048	2,829,474	0.820	0.537	0.558


Round 2: solve_rate on the near-trivial end, by config
mean_o1_solve_rate (k=0..4 uncolored nodes)
1.0
0.8
0.6
0.4
0.2
0.0
L=8
ff=1024
L=8
ff=2048
L=10
ff=1024
L=10
ff=2048
L=12
ff=1024
L=12
ff=2048


solve_rate across k=0-4 uncolored nodes, top configs vs. baseline
solve_rate
1.0
0.8
0.6
0.4
0.2
0.0
k=0
k=1
k=2
k=3
k=4
k = number of uncolored nodes (fixed small constant, n_nodes=24)
L=12, ff=1024
L=8, ff=1024
L=10, ff=1024
baseline L=2, ff=256 (150 epochs)


Progression: baseline -> round 1 -> round 2
mean_o1_solve_rate (near-trivial end)
agg_solve_rate (full rho range)
1.0
0.8
0.6
0.4
0.2
0.0
0.48
0.08
0.90
0.49
0.98
0.72
baseline
(L=2, ff=256,
150 epochs)
round 1 best
(L=8, ff=1024,
150 epochs)
round 2 best
(L=12, ff=1024,
300 epochs)

Takeaways

- Depth was the dominant lever, but only paired with enough width: at ff=128, even num_layers=8 only reached 0.698; num_layers=12 at dim_feedforward=1024 reaches 0.980.

- dim_feedforward=2048 underperformed 1024 at every depth tested in round 2 (e.g. L=12: 0.980 at ff=1024 vs 0.870 at ff=2048) -- likely undertrained at 300 epochs, not a real capacity loss.

- Recommended config: d_model=64, nhead=4, num_layers=8-12, dim_feedforward=1024, epochs=300, n_train_instances=5000.

- Full rho-range solve_rate (agg_solve_rate) also improved a lot (0.005 -> 0.72), but stays well below the near-trivial end's ~0.98 -- the harder/deeper reasoning chains remain a separate, still-open problem.